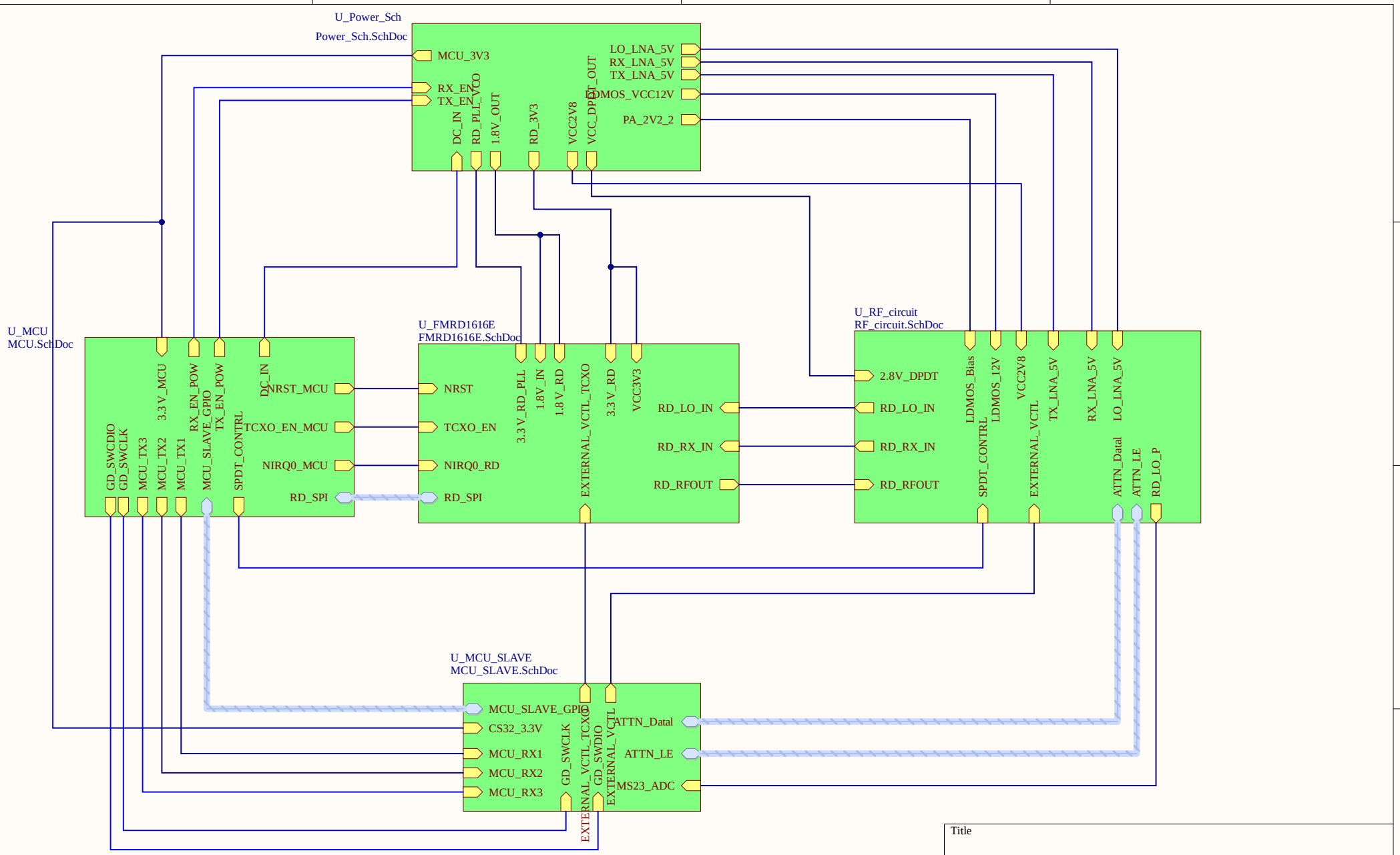
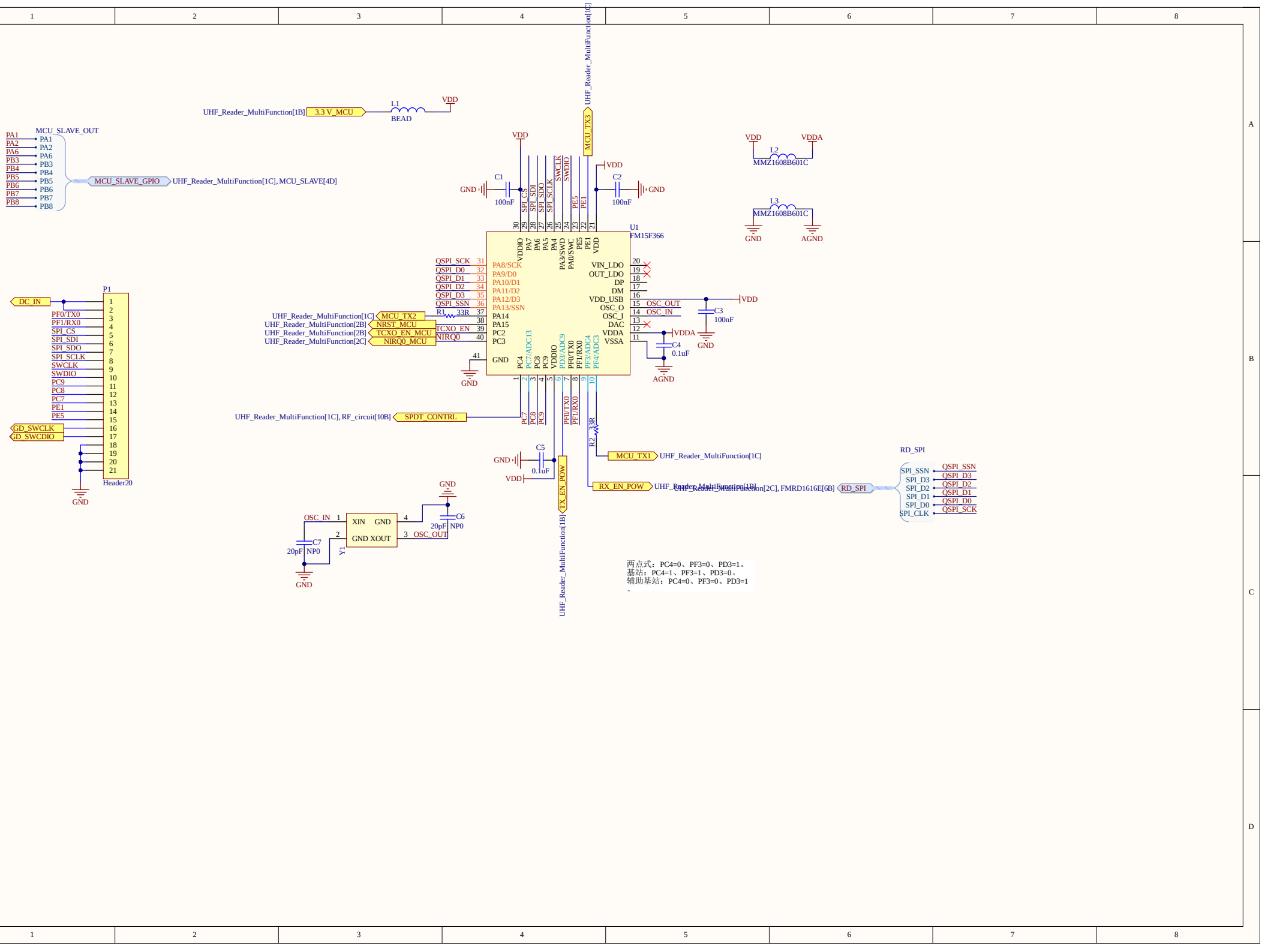
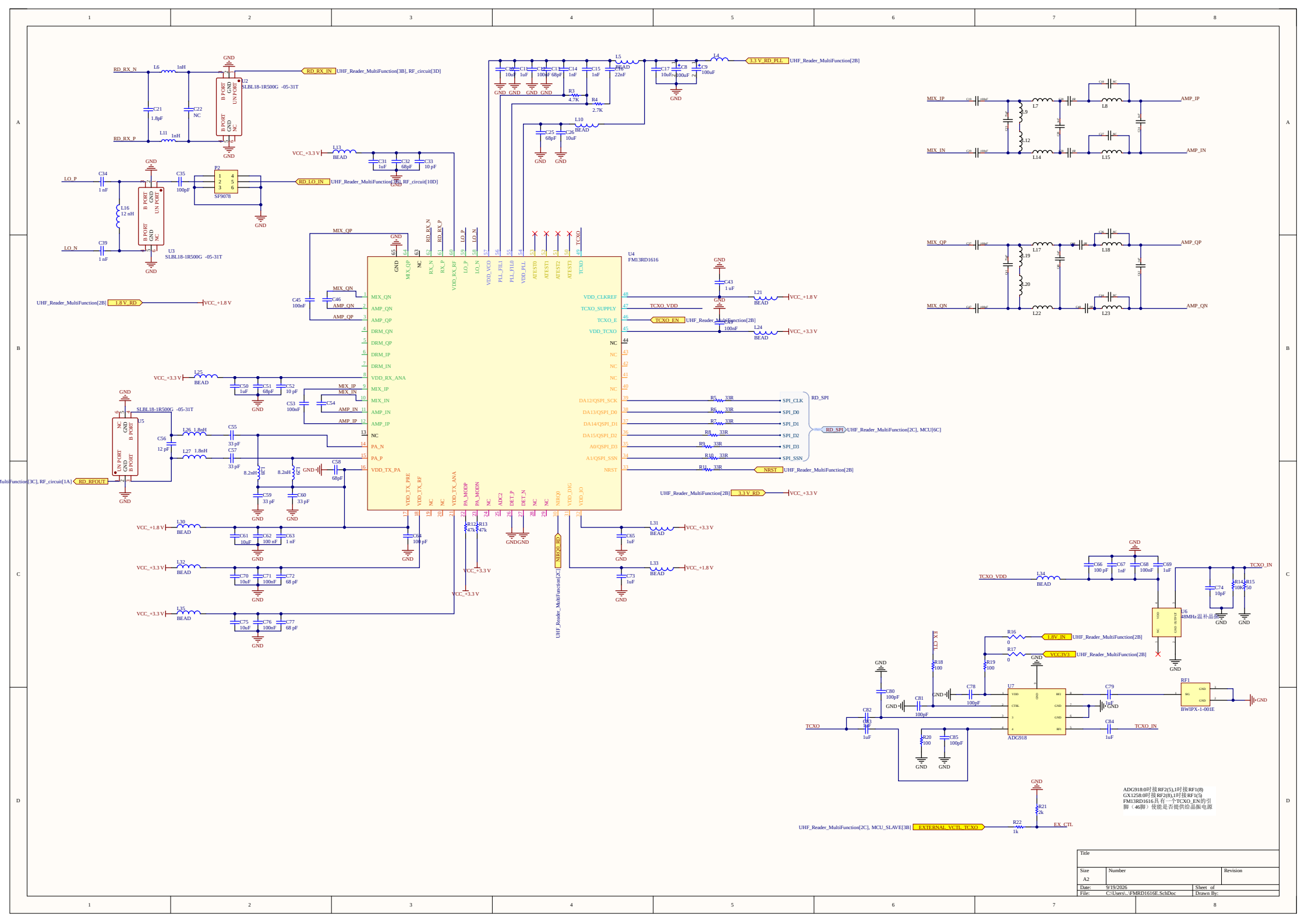
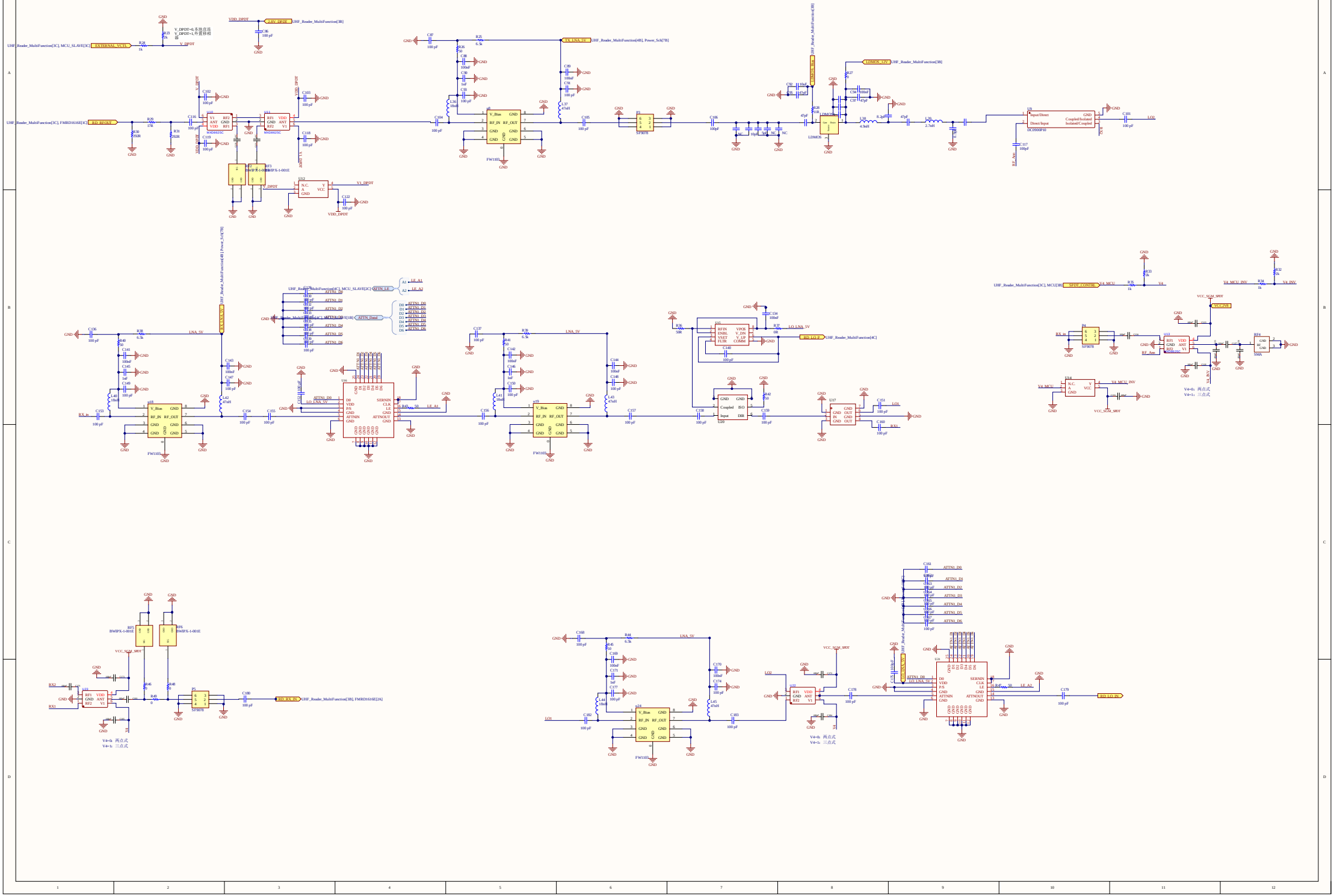
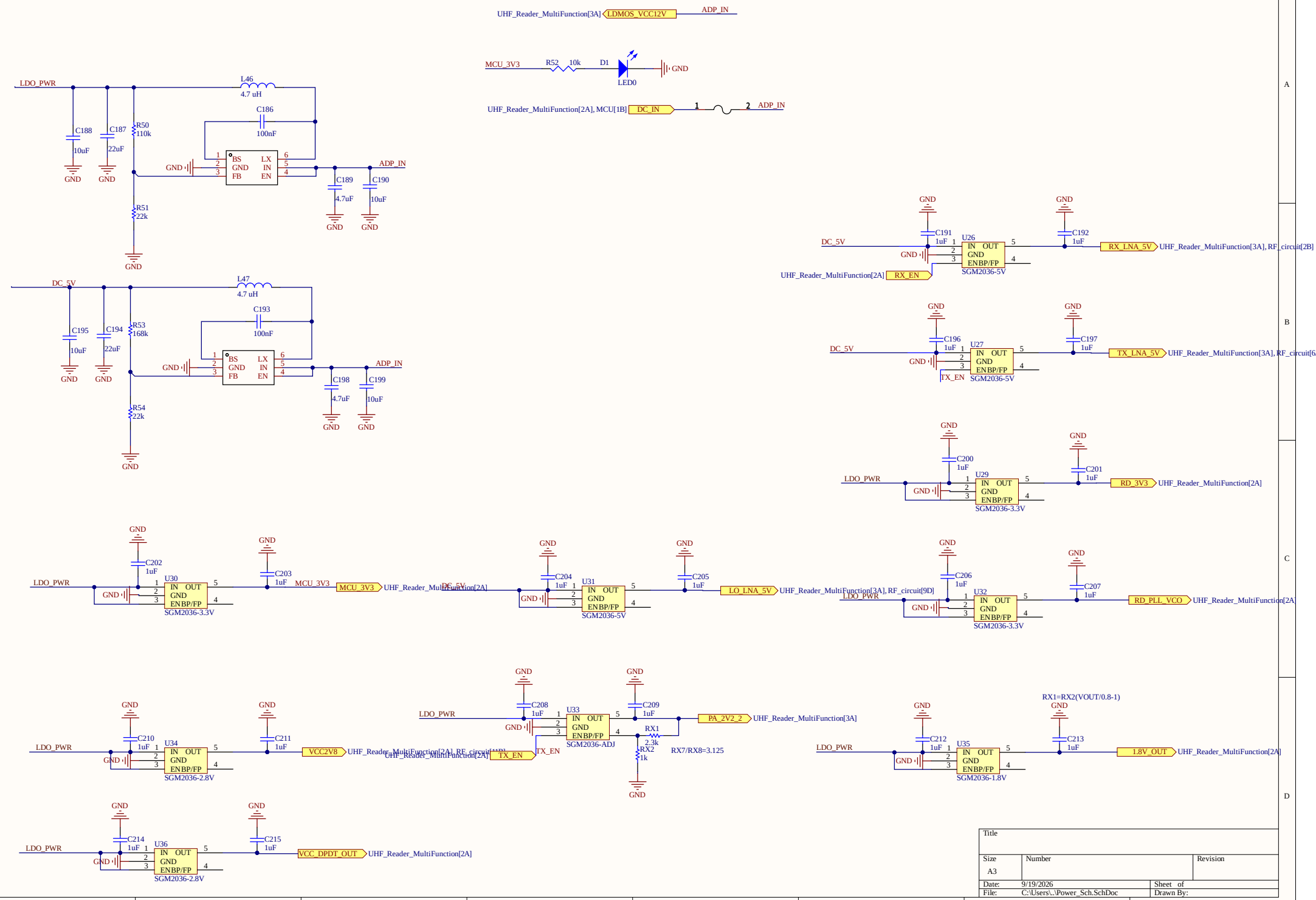


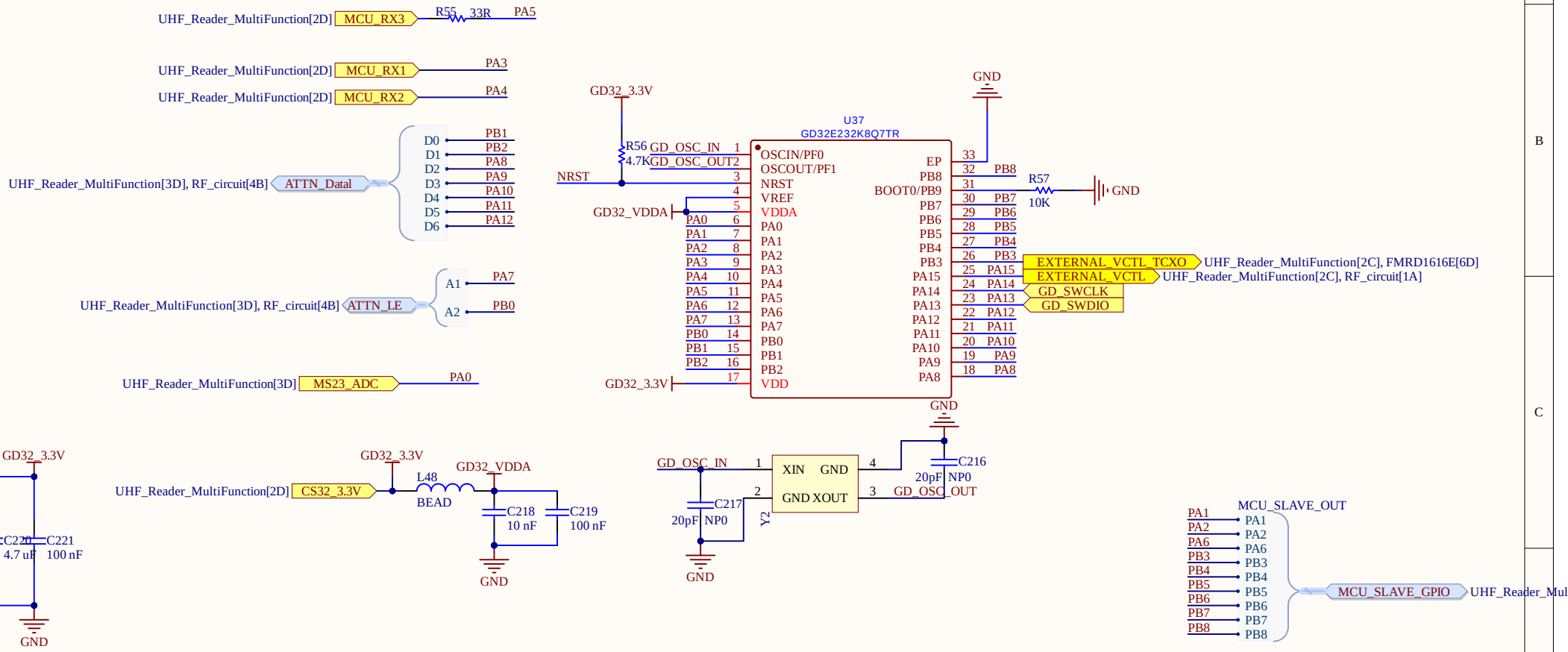
Title		
Size: A4	Number	Revision
Date: 9/19/2026	Sheet of	
File: C:\Users\..\UHF_Reader_MultiFunction.SchDoc	Drawn By:	


UHF_Reader_MultiFunction[1B]
3.3 V_MCU
L1
BEAD
VDD
UHF_Reader_MultiFunction[1C]
MCU_TX3
MCU_SLAVE_OUT
PA1
PA2
PA6
PB3
PB4
PB5
PB6
PB7
PB8
MCU_SLAVE_GPIO
UHF_Reader_MultiFunction[1C], MCU_SLAVE[4D]
C1
100nF
C2
100nF
GND
VDDA
L2
MMZ1608B601C
L3
MMZ1608B601C
AGND
U1
FM15F366
SPI_CS
SPI_SDI
SPI_SDO
SPI_SCLK
SWCLK
SWDIO
PE5
PE1
VDDIO
PA7
PA6
PA5
PA4
PA3/SWD
PA0/SWC
VDD
QSPI_SCK
QSPI_D0
QSPI_D1
QSPI_D2
QSPI_D3
QSPI_SSN
PA8/SCK
PA9/D0
PA10/D1
PA11/D2
PA12/D3
PA13/SSN
PA14
PA15
PC2
PC3
GND
VIN_LDO
OUT_LDO
DP
DM
VDD_USB
OSC_O
OSC_I
DAC
VDDA
VSSA
OSC_OUT
OSC_IN
C3
100nF
C4
0.1uF
R1
33R
MCU_TX2
NRST_MCU
TCXO_EN_MCU
NIRQ0_MCU
TCXO_EN
NIRQ0
PC7/ADC13
PC8
PC9
VDDIO
PD3/ADC9
PF0/TX0
PF1/RX0
PF3/ADC4
PF4/ADC3
SPDT_CONTRL
UHF_Reader_MultiFunction[1C], RF_circuit[10B]
C5
0.1uF
R2
33R
MCU_TX1
RX_EN_POW
TX_EN_POW
UHF_Reader_MultiFunction[1B]
P1
DC_IN
PF0/TX0
PF1/RX0
SPI_CS
SPI_SDI
SPI_SDO
SPI_SCLK
SWCLK
SWDIO
PC9
PC8
PC7
PE1
PE5
GD_SWCLK
GD_SWCDIO
Header20
OSC_IN
XIN
GND
GND XOUT
OSC_OUT
C6
20pF
NP0
C7
20pF
NP0
Y1
RD_SPI
SPI_SSN
SPI_D3
SPI_D2
SPI_D1
SPI_D0
SPI_CLK
QSPI_SSN
QSPI_D3
QSPI_D2
QSPI_D1
QSPI_D0
QSPI_SCK
UHF_Reader_MultiFunction[2C], FMRD1616E[6B]
两点式：PC4=0、PF3=0、PD3=1。
基站：PC4=1、PF3=1、PD3=0。
辅助基站：PC4=0、PF3=0、PD3=1

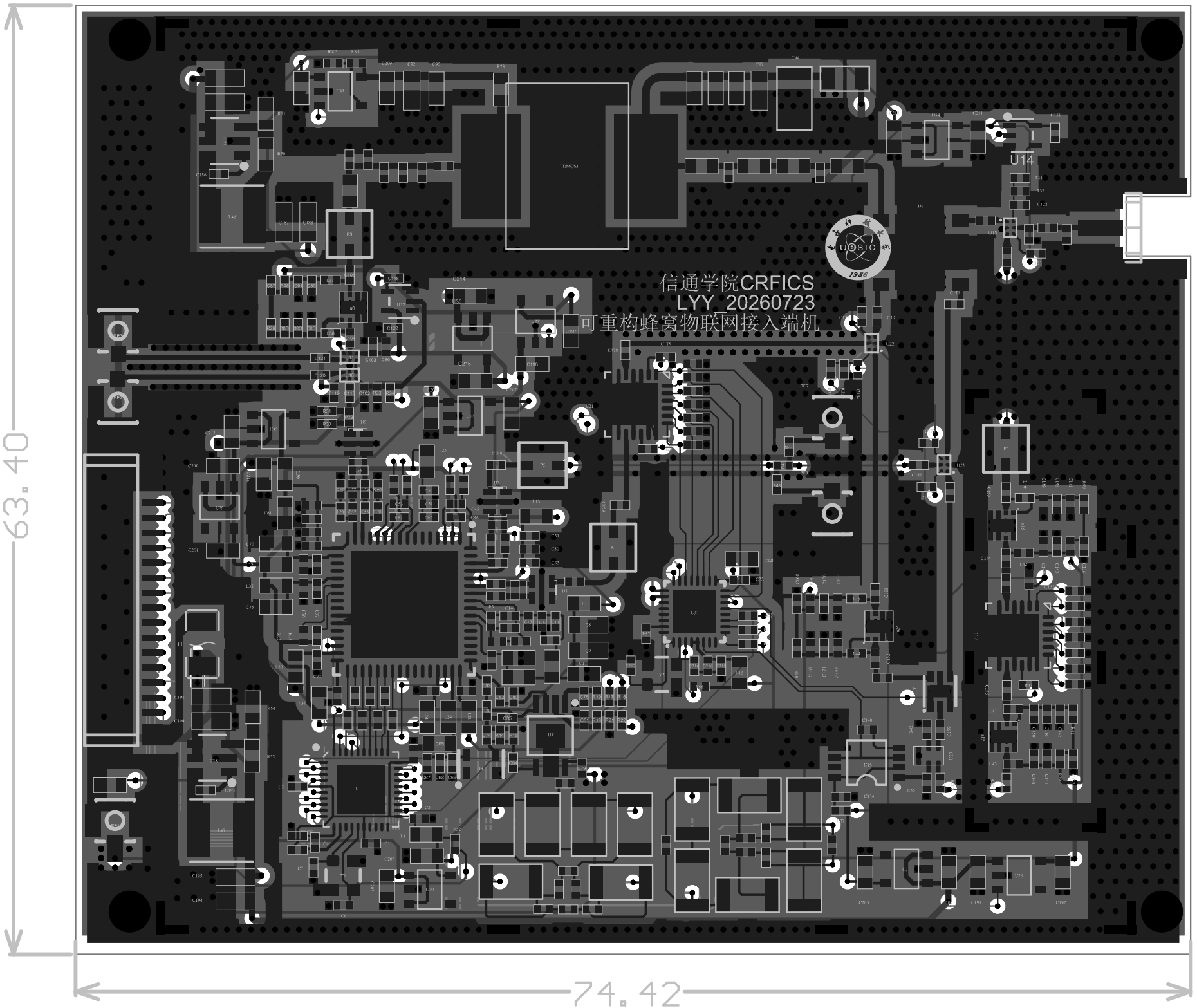

RD_RX_N
RD_RX_P
RD_RX_IN
UHF_Reader_MultiFunction[3B], RF_circuit[3D]
SLBL18-1R500G -05-31T
LO_P
LO_N
RD_LO_IN
UHF_Reader_MultiFunction[3B], RF_circuit[10D]
SF9078
MIX_QP
MIX_QN
AMP_QP
AMP_QN
MIX_IP
MIX_IN
AMP_IP
AMP_IN
FM13RD1616
3.3 V_RD_PLL
UHF_Reader_MultiFunction[2B]
1.8 V_RD
VCC_+1.8 V
VCC_+3.3 V
TCXO_VDD
TCXO_EN
RD_SPI
SPI_CLK
SPI_D0
SPI_D1
SPI_D2
SPI_D3
SPI_SSN
NRST
UHF_Reader_MultiFunction[2C], MCU[6C]
UHF_Reader_MultiFunction[3C], RF_circuit[1A]
RD_RFOUT
NIRQO_RD
ADG918
RF1
BWIPX-1-001E
TCXO
TCXO_IN
EXTERNAL_VCTL_TCXO
EX_CTL
UHF_Reader_MultiFunction[2C], MCU_SLAVE[3B]
ADG918:0时接RF2(5),1时接RF1(8)
GX1258:0时接RF2(8),1时接RF1(5)
FM13RD1616具有一个TCXO_EN的引脚（46脚）使能是否提供给晶振电源
Title
Size
A2
Number
Revision
Date: 9/19/2026
File: C:\Users\...\FMRD1616E.SchDoc
Sheet of
Drawn By:




UHF_Reader_MultiFunction[3A]
LDMOS_VCC12V
ADP_IN
MCU_3V3
R52
10k
D1
GND
LED0
UHF_Reader_MultiFunction[2A], MCU[1B]
DC_IN
ADP_IN
LDO_PWR
L46
4.7 uH
C186
100nF
C188
10uF
C187
22uF
R50
110k
R51
22k
BS
GND
FB
LX
IN
EN
C189
4.7uF
C190
10uF
DC_5V
L47
4.7 uH
C193
100nF
C195
10uF
C194
22uF
R53
168k
R54
22k
C198
4.7uF
C199
10uF
C191
1uF
U26
SGM2036-5V
C192
1uF
RX_LNA_5V
UHF_Reader_MultiFunction[3A], RF_circuit[2B]
RX_EN
C196
1uF
U27
C197
1uF
TX_LNA_5V
TX_EN
C200
1uF
U29
SGM2036-3.3V
C201
1uF
RD_3V3
UHF_Reader_MultiFunction[2A]
C202
1uF
U30
C203
1uF
C204
1uF
U31
SGM2036-5V
C205
1uF
LO_LNA_5V
UHF_Reader_MultiFunction[3A], RF_circuit[9D]
C206
1uF
U32
C207
1uF
RD_PLL_VCO
C210
1uF
U34
SGM2036-2.8V
C211
1uF
VCC2V8
UHF_Reader_MultiFunction[2A], RF_circuit[4B]
C208
1uF
U33
SGM2036-ADJ
C209
1uF
PA_2V2_2
RX1
2.3k
RX2
1k
RX7/RX8=3.125
C212
1uF
U35
SGM2036-1.8V
C213
1uF
1.8V_OUT
RX1=RX2(VOUT/0.8-1)
C214
1uF
U36
C215
1uF
VCC_DPDT_OUT
IN
OUT
GND
EN
BP/FP
Title
Size
A3
Number
Revision
Date:
9/19/2026
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File:
C:\Users\..\Power_Sch.SchDoc
Drawn By:



Title		
Size A4	Number	Revision
Date: 9/19/2026		Sheet of
File: C:\Users\..\MCU_SLAVE.SchDoc		Drawn By:


63.40
74.42
信通学院CRFICS
LYY 20260723
可重构蜂窝物联网接入端机
U14